

Tutorial-5

An Email Service provider wants to classify emails as Spam (0) or Not Spam using a linear sum classifier.

Input Feature:

- No. of suspicious words.
- No. of external links.
- Attachment present (0/1)

Target Variable: Class (Spam = 1, Not Spam = 0)

x_1	x_2	x_3	y
6	3	1	-1
1	0	0	-1
4	2	1	+1
0	0	0	-1
5	2	1	+1

Unit weights: $w_1 = w_2 = w_3 = 1$, $b = 0$, $\eta = 1$

Sample 1: (6, 3, 1) $y = -1$

$$f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

$$f(n) = 6(1) + 3(1) + 1(1) + 0 \\ = 10$$

$$y_i(f(n)) = y_i(f(n)) \Rightarrow (-1)(10) = -10$$

If $y[f(n)] \geq 1 \rightarrow$ Perfect Classification.

$y[f(n)] < 1 \rightarrow$ Miss Classification

— here, $y(f(n)) < 1$, So Miss Classification

Updating weights & Bias.

$$w_0^{\text{new}} = w_0^{\text{old}} + \eta \sum_{i=1}^n c_i y_i \quad [C=1]$$

$$w_0^{\text{new}} = 0 + 1 \cdot (1) \cdot (-1) \Rightarrow -1$$

$$w_1^{\text{new}} = w_1^{\text{old}} + \eta \left(w_1^{\text{old}} - \sum_{i=1}^n c_i y_i x_i \right)$$

$$= 1 - 1 \left[1 - 1 \cdot (-1) \cdot (6) \right] \Rightarrow -6$$

$$w_2^{\text{new}} = 1 - 1 \left[1 - 1 \cdot (-1) \cdot (3) \right] \Rightarrow -3$$

$$w_3^{\text{new}} = 1 - 1 \left[1 - 1 \cdot (1) \cdot (1) \right] \Rightarrow -1$$

$$(w_0, w_1, w_2, w_3) = (-1, -6, -3, -1)$$

Sample 2. (1, 0, 0)

$$f(x) = (-6)(1) + (-3)(0) + (-1)(0) + (-1) = -7$$

$$y(f(x)) = (-1)(-7) = 7$$

$$y(f(x)) \geq 1 \rightarrow \text{Perfect Classification.}$$

Update Weight:

$$w_0^{\text{new}} = (-6) - (1)(-6) \Rightarrow 0$$

$$w_1^{\text{new}} = (-3) - (1)(-3) \Rightarrow 0$$

$$w_3^{\text{new}} = (-1) - (1)(-1) \Rightarrow 0$$

$$\boxed{w_0^{\text{new}} = w_0^{\text{old}}}$$

Sample 3: (0, 1, 0)

$$f(x) =$$

y

update

Sample 4

Upd

Sample 3: $(4, 2, 1)$ $[y=1]$ $w_1, w_2, w_3, w_0 = (0, 0, 0, -1)$

$$f(x) = 0 \times 4 + 0(2) + 0(1) + (-1) = -1$$

$$y[f(x)] = (-1)(1) = -1$$

$y[f(x)] < 1 \rightarrow$ Missclassification.

Update weights & Bias

$$w_1^{\text{new}} = 0 - 1(0 - 1(1)(4)) = 4$$

$$w_2^{\text{new}} = 0 - 1[0 - 1(1)(2)] = 2$$

$$w_3^{\text{new}} = 0 - 1[0 - 1(1)(1)] = 1$$

$$w_0^{\text{new}} = -4 + 1(1) = -3$$

Sample 4 $(0, 0, 0)$ $[y=1]$ $w_1, w_2, w_3, w_0 = (4, 2, 1, -3)$

$$f(x) = 4(0) + 2(0) + 1(0) + (-3) = -3$$

$$y[f(x)] = -1 \times -3 = 3$$

$y[f(x)] < 1 \rightarrow$ Missclassification

Updating wts & Bias

$$w_1^{\text{new}} = 4 - 1[4 - 1(-1)(0)] = 0$$

$$w_2^{\text{new}} = 2 - 1[2 - 1(-1)(0)] = 0$$

$$w_3^{\text{new}} = 1 - 1[1 - 1(-1)(0)] = 0$$

$$w_0 = 0 + 1(-1) = -1$$

Sample 5: $(5, 2, 1)$ $(w_0, w_1, w_2, w_3) \rightarrow (-1, 0, 0, 0)$

$$f(x) = 0(5) + 0(2) + 0(1) + (-1) \Rightarrow -1$$

$y(f(x)) < 1 \rightarrow$ Miss Classification

Updating weights & Bias:

$$w_0^{\text{new}} = 0 - 1 [0 - 1(1)5] = 5$$

$$w_1^{\text{new}} = 0 - 1 [0 - 1(1)2] = 2$$

$$w_2^{\text{new}} = 0 - 1 [0 - 1(1)1] = 1$$

$$w_3^{\text{new}} = -1 + 1(1)(1) = 0$$

Final weights & Bias are

$$w_0 = 5, w_1 = 2, w_2 = 1, w_3 = 0$$

$$\text{Margin} = \frac{2}{\|W\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2}}$$

$$\Rightarrow \frac{2}{\sqrt{(5)^2 + (2)^2 + (1)^2}} = \frac{2}{\sqrt{30}} \approx 0.365$$